Easing Elevator Congestion in a Vertical Classroom Building: A Simulation Study of the Mundelein Center

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October 2026

Abstract

Mundelein Center, on Loyola University Chicago's Lakeshore Campus, holds 54 general-purpose classrooms on twelve floors but is served by only two elevator banks: four cars that reach floor 9 and two cars that reach floor 14. Each class changeover produces long lines. We build a discrete-time simulation of one 15-minute passing period to compare two low-cost behavioral measures, holding backpacks in front and walking to and from floors 2 and 3, and to test whether running floors 10–14 on a time offset spreads the load. In the baseline, mean waiting time is 5.2 minutes and 23% of students arrive late. Backpacks cut mean wait by 31%; walking cuts it by only 6%; together they cut it by 35% and lateness to 11%. Offsetting the upper floors shortens the time to clear the queue by about 20% but, combined with the other measures, *raises* the share of late students.

These results are conditional on the model in two ways. The parameters are estimates, not measurements. More consequentially, the simulator embeds fixed behavioral and operational assumptions (for example, that students never abandon a queue and that the controller treats all cars alike), which the sensitivity analyses cannot test. We list them, state which conclusions are most exposed, and treat absolute minutes as outputs of one model rather than facts about Mundelein Center.

1 Introduction

Mundelein Center is a tall, narrow classroom building. Its classrooms are spread from floor 1 to floor 14, and all of them are reached through two elevator banks of four and two cars. Only the two-car bank travels above floor 9. Every 75 minutes on Tuesdays and Thursdays and every 50 minutes on Mondays, Wednesdays, and Fridays, students leaving one class and arriving for the next converge on the elevators. Cars leave the upper floors carrying four to six people and still stop at every floor below, even though they are physically full, so students on the lower floors wait for cars that cannot take them.

We focus on classes that end before 4 PM and on two behavioral measures that cost nothing to implement: asking students to take their backpacks off and hold them in front (our estimate: two more riders per car), and asking students on floors 2 and 3 to walk. A third question is scheduling: because the registrar's grid fixes the class times, can the upper floors run on a shifted clock so that their peak does not coincide with the rest of the building?

The contribution is a transparent, reproducible model, not a measurement; the classical treatments of elevator traffic [2, 3] motivate its structure. All code, parameters, and result tables accompany this paper (Section 9).

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A reader should keep two kinds of uncertainty apart. The first is *parametric*: elevator speed, door time, boarding time per rider, the number of extra riders gained by holding backpacks, and the compliance rates are estimates. We can vary them, and we do. The second is *structural*: the simulator encodes a particular account of how students behave in a queue and how the elevators operate, and that account is not a parameter that can be swept. For example, the model gives students no way to give up on the line, represents partial compliance as a coin flip per car, and fills every seat of every class. A structural assumption can change a result as much as any parameter we vary, but it will not show up in a sensitivity table, because it is the same in every run. Section 3.5 lists these assumptions, and the results, discussion, and recommendations are written with them in view.

We also state what kind of contribution this is. The paper presents a simulation framework and preliminary estimates of the effect of two measures. Its conclusions are model results, that is, what follows under stated assumptions. Some are robust to the parameters we varied, but none is yet an empirical prediction for the building, and the paper is not a basis for operational change. Section 8 ranks the work needed to move from model results toward evidence.

2 Background

2.1 The building and its classrooms

We counted classrooms from the university's classroom A/V listing for Mundelein Center. Table 1 gives the count per floor. There are no classrooms on floors 8 and 9, so the one-bank limit at floor 9 splits the building cleanly into a lower group (floors 1–7, 45 rooms) and an upper group (floors 10–14, 9 rooms) that only the two-car bank can reach.

Table 1: Classrooms per floor in Mundelein Center.

Floor	1	2	3	4	5	6	7	10	11	12	13	14
Rooms	2	4	5	8	10	11	5	4	1	1	1	2

Floor 1 is an atrium lobby, and the stairs from floor 1 to floor 2 feel like two ordinary flights. Students leave the building through the lobby.

2.2 The scheduling grid

The Registrar's standard grid [1] defines seven 50-minute Monday–Wednesday–Friday blocks (8:15 to 3:35) and five 75-minute Tuesday–Thursday blocks (8:30 to 3:45) before the evening blocks, which start at 4:15 and are outside the scope of this study. Every consecutive pair of blocks, in either pattern, is separated by a 15-minute passing period. Our simulation models that period.

Classes seat 30–35 students, but enrollment is often 15–25. Elevator cars are rated for 3,000 lb, which is roughly 17–20 riders by weight; observed loads are four to six, so space, not weight, is the binding limit.

3 Model

3.1 Demand

One run simulates a single passing period. Time zero is the nominal end of the outgoing class and $T = 900$ seconds is the nominal start of the incoming class. Each classroom holds an outgoing and

an incoming class independently with probability p_{occ}; each class has between 15 and 25 students, uniformly at random. Leavers depart uniformly over the first two minutes. A fraction q of leavers go on to another class in the building (assigned to an incoming room with probability proportional to remaining seats, and to an adjacent floor with probability a); the rest leave via the lobby. The seats in incoming rooms that transfers do not fill are occupied by students who arrive from the lobby with a triangular distribution between ten and one minutes before class start, peaking three minutes before. Every student is ready at an elevator the moment they leave their room or enter the lobby.

3.2 Elevators

Cars are simulated in one-second steps. The four-car bank serves floors 1–9 and the two-car bank floors 1–14. Passengers whose trip includes a floor above 9 can use only the two-car bank. Cars follow a collective rule: continue in the current direction while there are riders to drop or calls ahead; otherwise reverse or go idle. Idle cars go to the nearest unclaimed call. Crucially, cars do *not* skip a floor when full: a car stops for any call in its direction of travel and then boards as many waiting people as it has room for. This builds in the empty stops observed in practice; how often they occur in the model (14.5% of stops in the baseline) is an output that we have not compared with observation. A car's capacity for a trip is drawn uniformly from $\{4, 5, 6\}$, and with backpack compliance b it is increased by two with probability b, independently for each car load.

Timing parameters are 3 seconds between floors (6 seconds from floor 1 to 2), 8 seconds of fixed door time per stop, and 1 second per rider boarding or alighting. These are estimates, not measurements. They are the *parameters* of the model. The rules above, and the demand rules of the previous subsection, are its *structure*, discussed in Section 3.5.

3.3 Behavioral measures and the offset

Walking. A rider whose trip starts on floor 2 or 3 and goes down walks with probability w_d (0.8 in the main experiments). A rider going up to floor 2 walks with probability 0.5 and to floor 3 with probability 0.2, reflecting the atrium. Walkers take 20 seconds per flight (40 for the atrium) and do not use the elevators. *Backpacks.* The compliance parameter b above, set to 1 in the main experiments. *Offset.* Classes on floors 10–14 start and end o minutes earlier than the grid ($o \in \{0, 5, 10\}$), preserving their length. A student moving from a lower floor to an upper floor therefore has $15 - o$ minutes between classes.

3.4 Metrics

For each run we record the mean and 90th-percentile wait (from when a rider is ready to when the rider boards), the share of students who reach their next class after it starts (walkers included), the share of stops at which nobody boards or alights, and the time to clear all queues. All figures average 60 replicates with different random seeds; we report standard errors of the mean for wait and lateness. Wait statistics cover only students who ride (walkers wait zero and are excluded), lateness includes walkers, and the clearing time is read on the registrar's clock, from the nominal end of the outgoing class, for both floor groups. The standard errors measure replicate-to-replicate variation *within the model*. They say nothing about whether the model is right.

3.5 Structural assumptions

Table 2 lists the assumptions that are built into the simulator's rules rather than exposed as parameters, with where each lives in the code and which results we judge most exposed to it. The last column is our

reasoning, not a test: none of these assumptions has been varied, so we do not know how large any of the effects is.

Table 2: Structural assumptions embedded in the simulator. None is varied in any experiment.

Assumption	What the simulator does	Results most exposed
Students never leave the queue	Every student who is not a designated walker waits for an elevator, however long the line. Nobody takes the stairs on their own, gives up, or leaves early (`split_walkers`). The walking measure is applied to a baseline in which no one walks.	Baseline wait and lateness; the size of the walking effect; the benefit of any measure that shortens lines.
Compliance is a per-car coin flip	Backpack compliance b adds exactly two seats to a car load with probability b, independently of the car, the floor, and the riders (`fresh_capacity`). Compliance is never clustered by class, floor, or time of day, and the bonus does not depend on the base capacity or on how many people are waiting.	The size of the backpack effect; its near-linearity in compliance; the ranking of backpacks over walking.
Space is the only constraint on boarding	Everyone in line boards if there is room; nobody declines a crowded car or waits for the next one. Boarding is first-in-first-out, and an empty car takes the direction of the first person in line (`serve_floor`).	Wait and the share of empty stops.
A stop costs only door time	Travel is a constant 3 s per floor with no acceleration. Everything else about stopping is folded into a fixed 8 s plus 1 s per rider (`t_door`, `t_pax`). The door-time sweep varies the value of this cost, not its form.	The cost of empty stops, hence the value of removing riders from lower floors; the gap between short and long trips.
Controller and queues	Collective rule; one shared queue per floor from which any eligible car draws; the two-car bank serves lower floors on the same terms as the four-car bank; no zoning, no priority for the upper floors. Cars start the period at random floors, with no backlog from the previous period (`World`, `depart`).	The finding that floor 10 is the worst floor; the offset results; absolute clearing times.
Demand is generated independently and in full	Outgoing and incoming occupancy of a room are independent draws with the same p_{occ}. Every seat of every incoming class is filled, and nobody is absent. Leavers depart uniformly in the first two minutes, with no early or late dismissal; non-transferring arrivals come only through the lobby on a triangular schedule; transfers choose a room in proportion to open seats (`build_demand`).	Queue size and shape; lateness; the effect of the offset.
Parameters tuned to the story	p_{occ} was chosen so that the baseline shows a long line without collapse, and the transfer share q and adjacent share a are not measured. That the baseline resembles the description is therefore not evidence that the model is right.	All absolute levels.

Assumption	What the simulator does	Results most exposed
Lateness is a hard threshold	A student is late if they arrive after the nominal class start; there is no grace period and no instructor discretion. Offset classes keep their length.	Lateness percentages; the offset trade-off.
Every changeover is the same	One 15-minute period, the same for the 50-minute and 75-minute patterns, and for every block of the day.	Any statement about a particular time of day.

Two points follow. First, the sensitivity analyses in Section 4.4 vary parameters inside this fixed structure. They show that the ordering of the measures is robust to occupancy, door time, and adjacent-floor traffic. They do not show that it would survive different student behavior or a different controller. Second, the effects are not all of one kind. Some assumptions, such as students never leaving the queue, make the baseline look worse and could shrink the apparent benefit of every measure if they were relaxed. Others, such as the coin-flip compliance, affect how much each measure achieves. We have not tested any of this, and we have not tried to guess its net direction.

4 Results

All numbers in this section are outputs of the model described above, including the assumptions in Table 2. We emphasize the relative effects of the measures; the absolute minutes and percentages are illustrative.

The main experiments use $p_{occ} = 0.4$ (about 570 elevator riders per changeover), $q = 0.6$, and $a = 0.2$. We chose p_{occ} so that the baseline reproduces the qualitative description of a long line before every class without collapse; this is a calibration to a description, not a validation (Table 2). Section 4.4 shows how the results change with occupancy.

4.1 Behavioral measures

Table 3: Policies without an offset ($o = 0$). Wait is mean ± standard error across replicates; late is the share of students arriving after class start. Absolute values are illustrative outputs of the model.

Policy	Mean wait (s)	P90 wait (min)	Late (%)	Empty stops (%)	Clear (min)
Baseline	314 ± 8	12.4	22.9 ± 1.4	14.5	25.0
Backpacks	217 ± 6	8.7	12.7 ± 1.2	9.3	20.2
Walking	294 ± 8	11.9	19.6 ± 1.3	13.9	24.6
Both	203 ± 6	8.5	11.0 ± 1.0	8.7	19.9

Table 3 and Figure 1 show the headline result. Holding backpacks in front reduces mean wait by 31% and nearly halves the share of late students. Walking is much weaker: it removes only about 7% of elevator riders, because floors 2 and 3 hold 9 of the 54 rooms and only partial compliance is realistic on the way up. Together the two measures reduce mean wait by 35%, to 3.4 minutes, and lateness from 22.9% to 11.0%. The effect of backpacks is nearly linear in compliance: each additional 25% of compliance removes roughly 20–30 seconds of mean wait (Figure 2). Part of this linearity is built in, because compliance enters the model as a per-car probability of two extra seats; see Section 3.5.

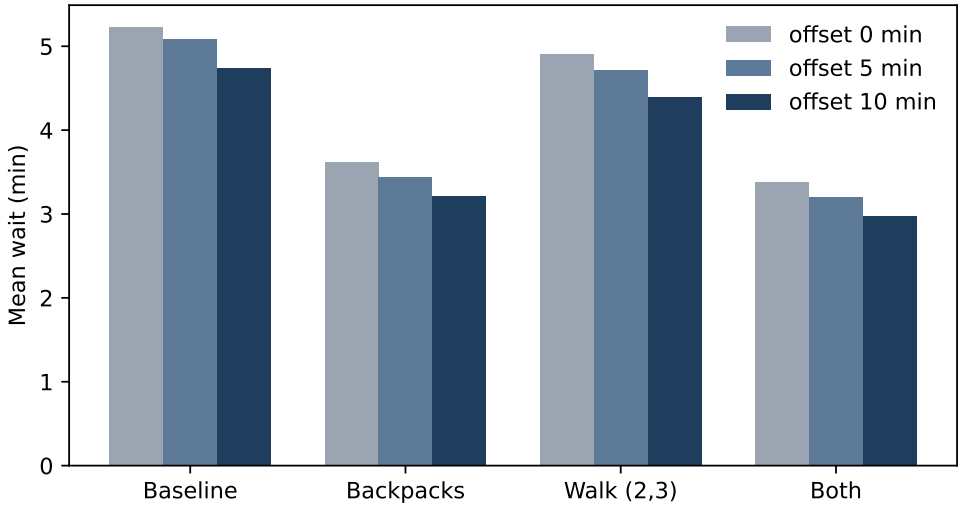


Figure 1: Mean wait by policy and offset of floors 10–14.

Figure 3 shows why. Immediately after class ends, about 300 people are waiting in the baseline and the queue drains slowly; the arriving students then add to a backlog that has not cleared when the next class starts. The combined measures lower the peak and, more importantly, speed up the drain.

4.2 Where the waiting happens

Waits are not uniform across the building (Figure 4). Floor 10, the first floor served only by the two-car bank, is the worst: riders there wait about 10 minutes in the baseline and 7.5 minutes with both measures. Floors 5–7 are next, at about 5–6 minutes. The measures help everywhere but do not remove the upper-floor bottleneck, which comes from the structure of the banks.

4.3 Time-multiplexing the upper floors

Table 4: Effect of running floors 10–14 o minutes early.

Policy	Offset (min)	Mean wait (s)	Late (%)	Clear (min)
Baseline	0	314	22.9	25.0
Baseline	5	305	24.2	22.3
Baseline	10	284	23.2	19.5
Both	0	203	11.0	19.9
Both	5	192	12.5	17.5
Both	10	179	14.4	15.9

These offset results are preliminary: a single changeover cannot capture how a shifted clock affects transfers across the whole teaching day (Section 8, item 6). An offset trades one problem for another (Table 4, Figures 1 and 5). A 10-minute offset shortens the mean wait by about 9% in the baseline and 12% with both measures, and shortens the time to clear the queue by 22% and 20%. It does not reduce lateness: in the baseline the change is within noise, and with both measures lateness *increases*

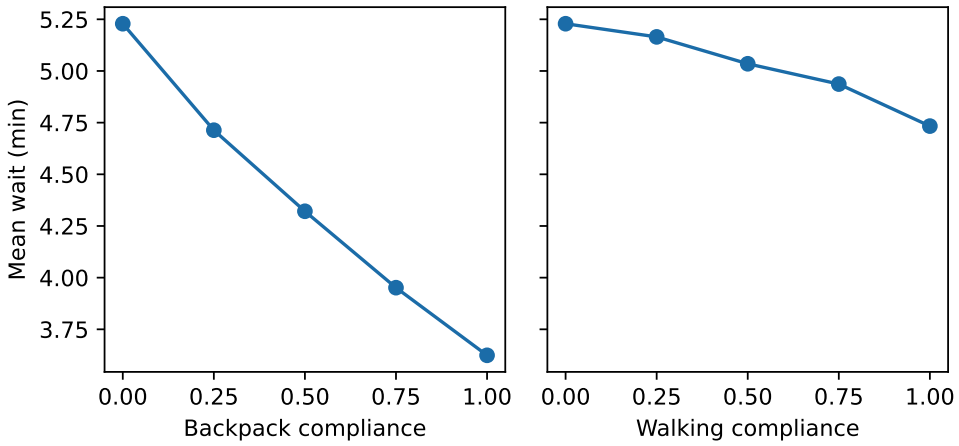


Figure 2: Mean wait as compliance grows. Left: backpacks. Right: walking (down-compliance shown; up-compliance scaled at 0.6 and 0.25 for floors 2 and 3).

from 11.0% to 14.4%, because students moving from a lower floor to an upper one now have only five minutes. Students moving between adjacent floors are unaffected, since the group boundary falls between floors 7 and 10, which have no classrooms in between; the cost falls entirely on students changing between the lower and upper groups. A 5-minute offset gives about half of the reduction in clearing time at about half of the added lateness, so we regard it as the more defensible choice if the registrar would consider off-grid start times.

4.4 Sensitivity to parameters

The sweeps below vary parameters only. The structural assumptions of Table 2 are the same in every run.

Occupancy. The absolute waits depend strongly on how many rooms are in use (Figure 6): from 2.6 minutes at 20% occupancy to 17 minutes at full occupancy in the baseline. The relative benefit of backpacks plus walking is stable, between 29% and 36% across the whole range. At 20% occupancy the baseline already has 7.9% late students, rising to 62% at full occupancy.

Door time. Doubling the door time per stop from 5 to 12 seconds raises the baseline mean wait from 236 to 432 seconds, while the relative benefit of both measures stays between 34% and 36%. Door time matters because every empty stop costs a full door cycle; this is the direct cost of the lack of a full-car bypass.

Adjacent-floor transfers. Varying the share of transfers that go to an adjacent floor from 10% to 30% changes the late share by only 1–2.5 points in either direction (Figure 7), because without stairs the adjacent-floor riders still use the elevators. Offering stairs for one-floor trips, an extension beyond the two measures requested, reduces the combined-policy mean wait from 196 to 176 seconds (about 10%) at an adjacent share of 30% and 80% stair use, and the late share from 10.4% to 9.1%.

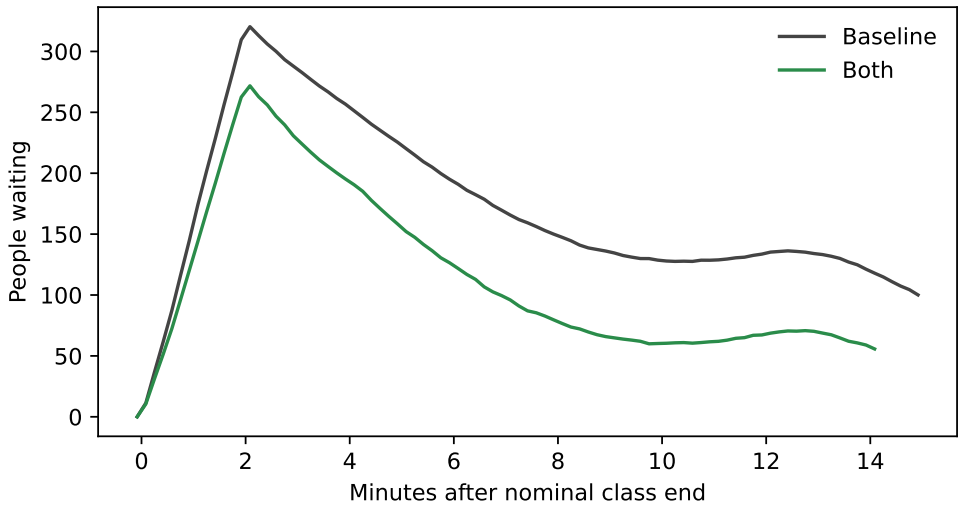


Figure 3: People waiting for an elevator over the passing period (mean of replicates, $o = 0$).

5 Discussion

What the results do and do not say. The model says that, *if* the structure of Table 2 holds and the parameters are roughly right, then relieving the space limit helps most, walking helps a little, and an offset trades queue length for lateness. It does not say how large any of these effects is at Mundelein Center, and it cannot say whether the structure holds. Parameter sweeps tell us which numbers matter within the model. Only observation can tell us whether the model is the right one.

Backpacks matter most. The cars leave full because of volume, not weight. Making room for two more riders per car is by far the largest single lever in the model, and it costs nothing but a message. That ranking depends on modelling choices: the benefit is a fixed two seats on every complying car, and students have no other way to adapt. Partial compliance gives partial benefit in the model, but this is partly by construction, since compliance is a per-car coin flip. In practice compliance may be clustered, for instance a few riders per car who do not comply may be enough to erase the extra room, and the model cannot show that.

Walking is a supporting measure. Floors 2 and 3 are only 17% of the classrooms, and the atrium makes the climb to floor 2 feel like two flights. The gain comes mostly from the way down: walkers remove the empty stops on floors 3 and 2 that full cars make on their way to the lobby.

The upper floors need structural help. Floor 10 is the worst floor even after both measures, because all of its traffic goes through the two-car bank. Behavioral measures soften the problem; moving some of the load off the upper floors in the room assignment, or offsetting their clock, would act on the cause. The offset is a trade-off, not a free gain. This conclusion depends on the controller assumptions: the model has the two-car bank serve lower floors on the same terms as the four-car bank, with no priority for the upper floors, and a real controller may allocate its cars differently.

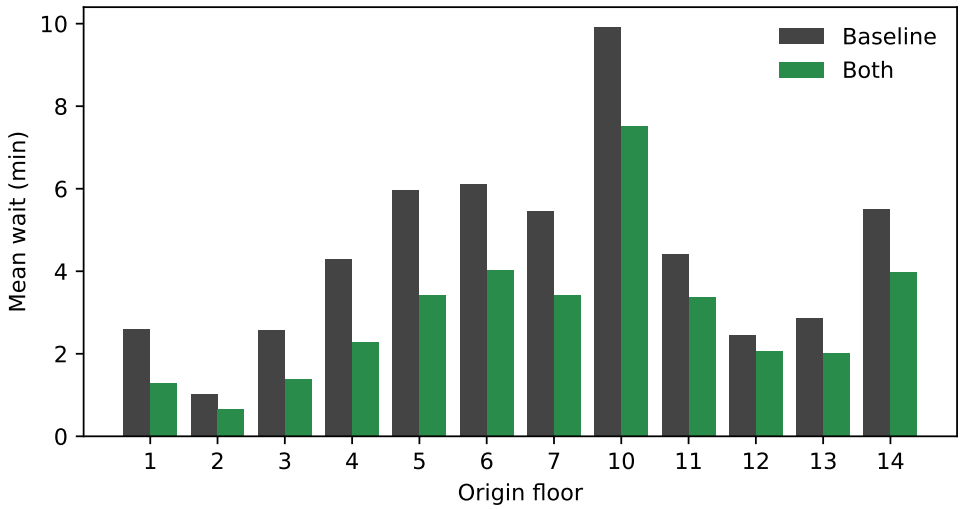


Figure 4: Mean wait by origin floor, baseline versus both measures ($o = 0$).

6 Limitations

Two kinds of assumption limit this study, and the second matters more than it may appear.

- **Parameters are estimates.** Every timing parameter (travel, door, boarding) and every behavioral share (leaving spread, arrival distribution, transfer share, compliance, the two extra seats per car) is an estimate, not a measurement. We varied several of them and the ordering of the measures held.
- **Structure is assumed, and was not varied.** The simulator fixes how students behave in a queue, how cars decide where to stop, how compliance is represented, and how demand is generated (Table 2). These choices are the same in every experiment, so no table in this paper measures their influence, and they may shape the results as much as the parameters we vary. The robustness reported in Section 4.4 should be read as robustness within one structure.
- **No validation.** We calibrated p_{occ} to match a qualitative description and compared nothing to observed queues, wait times, or stop counts. Agreement with the description is not evidence that the structure is right. The standard errors reflect replicate noise, not model error.
- **Relative results only.** The model produces comparisons between measures that we believe are reasonable, conditional on the points above. Absolute minutes and percentages should not be quoted as facts about Mundelein Center.
- **Unserved riders.** Earlier runs ended at 3,600 s and dropped riders still waiting, which understated congestion at the highest occupancies. The cutoff is now a backstop that no scenario reaches, unserved riders would count as late, and unserved counts are reported (Section 8, item 2). The accounting itself has not yet been tested independently.
- **Comparisons are not tested.** Measures are compared by their per-scenario means and standard errors, not by confidence intervals on paired differences, and no outcome other than the ones listed in the Metrics subsection is reported.

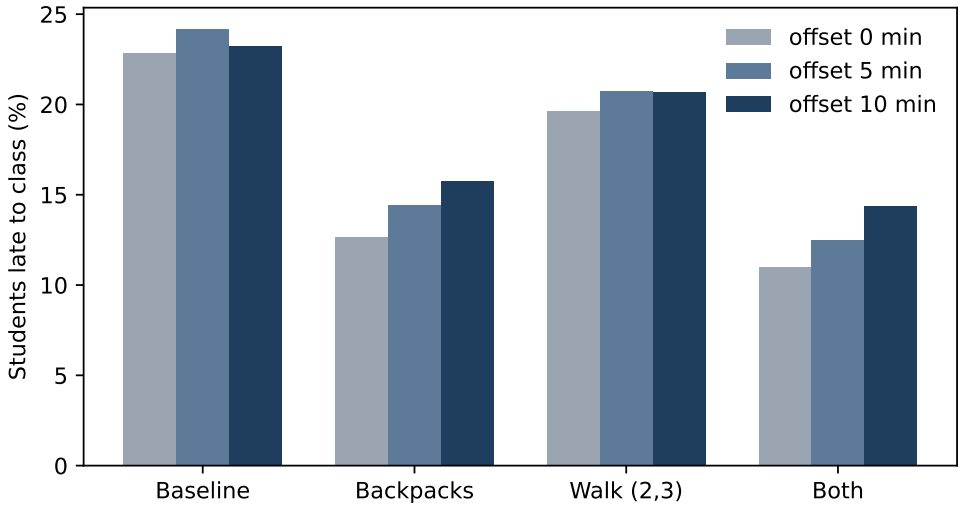


Figure 5: Share of students arriving after class start, by policy and offset.

- **One passing period.** We simulate one changeover from an empty building. Backlog from the previous changeover would make the baseline worse.
- **Uniform schedule.** Without a room-by-block schedule we assume every room is in use with probability p_{occ} in every block and cannot distinguish heavy blocks from light ones.
- **Simple dispatching.** The real controller is proprietary and may behave differently. We deliberately did not model a full-car bypass or down-peak mode, because these were out of scope.
- **Static students.** Students do not adapt by leaving earlier, taking the stairs, or skipping the line when they see it. This is one of the structural assumptions with the clearest real-world counterpart, since adaptation would reduce both waits and lateness in the baseline.

7 Conclusions and recommendations

The recommendations below are conditional on the model's parameters and on the structural assumptions in Table 2. We state them as what the model supports, not as predictions for Mundelein Center.

1. Ask students to hold backpacks in front. In the model this is the largest single improvement: about 31% less mean wait and half the late arrivals at full compliance.
2. Ask students on floors 2 and 3 to walk, especially downward. It is a smaller gain, about 6% of wait on its own, but it combines with the first measure, and the two together reduce mean wait by about 35%.
3. Treat the offset for floors 10–14 as a preliminary result, not a recommendation. A 5-minute offset shortens the time to clear the queue, but it costs students who move between floor groups, and a single-changeover model cannot show its effect on the rest of the teaching day (Section 8, item 6).
4. Do not expect behavioral measures alone to fix floor 10. In the model its waiting time is structural, subject to the controller caveat above.

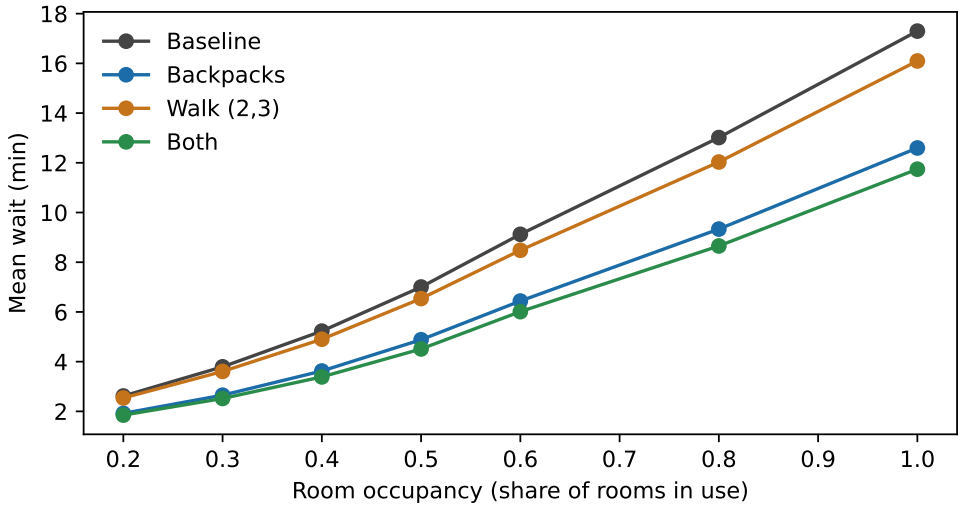


Figure 6: Mean wait as room occupancy grows.

5. Before quoting any number, validate the model against observation. The ranked agenda in Section 8 begins with a small observational study, independent tests of the simulator, and sensitivity and structural-robustness experiments; none has been done.

8 Future work

The next step is to shift from modeling to validation. The simulator already has enough detail to produce interesting results and it remains to be seen whether those results are meaningful. We therefore rank the work below by how much it would strengthen this project, and defer additions such as a full-day simulator, optimization of room assignments, and detailed pedestrian modelling to separate projects. None of the items has been done unless stated.

1. **Validate against observation.** Record 5–10 class changeovers: elevator arrival intervals, boarding and alighting counts, car occupancy, queue lengths, and waits at selected floors. Split the data into calibration changeovers, used to choose parameters, and validation changeovers, used only to test whether the calibrated model predicts congestion it has not seen. This also replaces the tuned p_{occ}. A small, controlled field experiment, subject to building approval and posted capacity limits, would compare car occupancy under ordinary behavior and under an explicit request to hold backpacks in front. It tests the assumption behind our strongest result: if the gain is about two riders, the model stands; if it is one, we rerun; if there is none, that is an equally useful finding.

2. **Verify the simulator independently of the experiments.** Add automated tests for passenger accounting, car capacity, movement, boarding rules, dispatch decisions, and termination, plus small deterministic scenarios that can be worked by hand (for example, one car, two floors, three passengers at fixed times). This includes one reporting defect that we have fixed. The simulator used to stop at 3,600 s and drop riders still waiting from the wait and lateness statistics. This never happened at $p_{occ} = 0.4$, but in the baseline at full occupancy up to 17 riders per replicate were dropped. The cutoff is now a 14,400 s backstop that no scenario reaches (the slowest clears

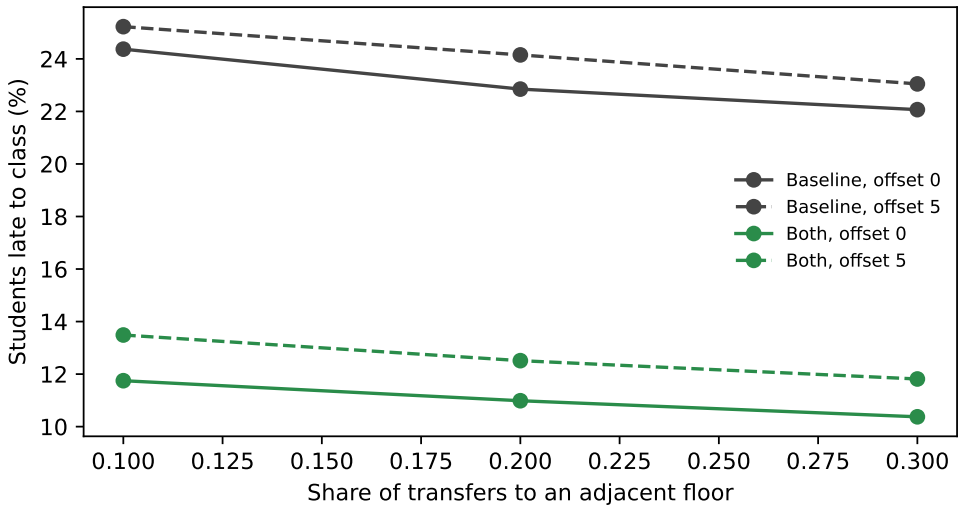


Figure 7: Share of late students against the adjacent-floor share of transfers.

in about 3,800 s), riders still waiting would count as late, and the unserved count is a column in every result table. Rerunning changed only two rows (the full-occupancy baseline and combined policy), and by less than 0.2 points of lateness; every other table is unchanged. The tests described above should confirm that this accounting is correct.

3. **Systematic parameter sensitivity, aimed at break-even points.** Repeat the experiments over a grid: extra seats from backpacks (0, 1, 2, 3), baseline capacity, occupancy, concentrated versus dispersed arrivals, and faster and slower boarding and door times. The goal is not more tables but the conditions under which a conclusion changes, for example the extra capacity below which the backpack measure no longer beats walking.

4. **Structural uncertainty.** Compare two or three plausible dispatch policies, alternative arrival and leaving behavior, and the other structural assumptions in Table 2 (students who leave the queue, per-rider compliance). We need not reverse-engineer the real controller, but we should show that the conclusions do not depend on an arbitrary implementation choice.

5. **Stronger statistical reporting.** Report confidence intervals for differences between measures. The scenarios already share random seeds, so the demand is common to all of them and paired differences are available, but this paper reports only per-scenario standard errors. Separate statistical from modelling uncertainty explicitly. Add outcomes beyond mean wait: total journey time, 90th- and 95th-percentile delay, on-time arrival, unserved counts, and results by floor and by bank, so that a gain on average is not bought with worse service on the upper floors.

6. **Evaluate scheduling offsets over several consecutive periods**, or remove them from the study. A start-time offset affects transfers throughout the day, so a single changeover cannot support a conclusion. Until then the offset results are preliminary.

7. **State the claims at the right level of evidence.** Distinguish model results (what follows under the stated assumptions), robust findings (what survives plausible alternatives), and empirical predictions (what we expect in the actual building). This paper is a simulation framework with preliminary estimates of the effect of two measures; it is not yet a basis for operational changes.

Almost all its conclusions are model results, only partly robust findings (ordering under three parameter sweeps), and not empirical predictions. This item is cheap and partly done in this version, by the structural-assumptions table and the reworded conclusions; it should be revisited as items 1–6 are completed.

If time is limited, the minimum for a substantially stronger paper is items 2, 1, 3–4, and 5, together with the claims in item 7.

9 Reproducibility

The simulator (`model/elevator_sim.py`), the experiment runner (`model/run_experiments.py`), all result tables (`results/*.csv`), and these figures are in the project repository. Running `python run_experiments.py` with 60 replicates regenerates every table and figure.

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